

Giwa Muhammad

Senior Full Stack Engineer

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Full Stack Engineer with 4+ years of experience architecting scalable SaaS platforms and distributed backend systems. Track record of full system ownership: architectural design, database modeling, API performance, caching strategy, asynchronous processing, and production deployment.

Specialized in React and Next.js frontend architecture and Node.js based microservices, with strong experience in Redis caching, event driven communication, database per service isolation, and multi tenant RBAC security models. Focused on scalability, reliability, and production grade system design.

CORE EXPERTISE

- Distributed Microservices Architecture
- Event-Driven Systems (RabbitMQ)
- Background Job & Queue Processing
- JWT Authentication & RBAC Models
- Scalable React / Next.js Architecture
- Multi-Tenant SaaS System Design
- Redis Caching & Latency Optimization
- Database Schema Design & Query Optimization
- Dockerized Infrastructure & CI/CD Pipelines
- Payments Integration (Stripe, Stripe Connect, Payoutstack)

PROFESSIONAL EXPERIENCE

CTO | SmartEdu

2026 - Present

- Lead product and engineering for a multi-tenant school-management SaaS platform serving schools across Africa.
- Architected a subdomain-per-tenant backend (Node.js, TypeORM, PostgreSQL) with Redis and BullMQ handling report generation, fee reminders, and payment reconciliation.
- Structured the frontend as a Turborepo of three Next.js applications sharing a component and design token library.
- Integrated Payoutstack for fee collection, including installment plans and a platform level revenue share.
- Own the go-to-market materials and school onboarding process alongside the technical roadmap.

Senior Full Stack Engineer, Architecture Lead | Raffles Energies

June 2024 - April 2025

- Led end-to-end architecture and backend development supporting web and mobile platforms.
- Architected and implemented the Node.js backend serving multiple client applications.
- Designed the relational database schema and indexing strategy from inception.
- Built the API layer serving both web and mobile clients under sustained production traffic.
- Implemented Redis caching and query optimization that measurably reduced response latency.
- Designed asynchronous job processing and workflow automation pipelines.
- Implemented JWT based authentication and tenant aware RBAC access control.
- Containerized services and managed production deployment workflows.
- Maintained high reliability through structured, low risk release cycles.

Backend Engineer | Centre for Innovation & Entrepreneurship

September 2023 - March 2024

- Scaled a Laravel based LMS platform supporting multi center operations for a government organization.
- Redesigned the relational schema to support expanded roles and institutional structure.
- Improved backend modularity and long term maintainability.
- Contributed to architectural evolution discussions.

Full Stack Engineer | Alphasol Limited

May 2023 - August 2023

- Led backend architecture and performance optimization for real time operational systems.
- Designed RESTful APIs supporting real time operational dashboards under continuous production load.
- Defined MongoDB and MySQL schema design and indexing strategies.
- Improved query performance significantly through indexing and load tuning.
- Implemented API caching strategies that reduced server load and improved throughput.
- Designed JWT authentication and role based authorization logic.
- Contributed to CI/CD workflows enabling bi-weekly production deployments.

- Migrated legacy Laravel Blade architecture into a structured Vue.js frontend system.
- Integrated the frontend with backend APIs and optimized data flow patterns.
- Reduced redundant API calls and improved UI performance.
- Participated in system level refactoring initiatives.

PROJECTS

StackMSP - Multi-Tenant Microservice SaaS Platform

Tech: Node.js, Go, PostgreSQL, Redis, RabbitMQ, Docker

- Designed a distributed system of isolated microservices, each with its own independent PostgreSQL database.
- Defined service boundaries and implemented event driven communication over RabbitMQ.
- Implemented a Redis caching layer to reduce API latency under load.
- Architected the system for high throughput production traffic with horizontal scaling in mind.
- Implemented a tenant scoped RBAC model and secure JWT authentication.
- Architected for high availability through service decoupling and fault isolation.

FairChild - Institutional Learning Management System

Tech: React, TypeScript, Node.js, Express, TypeORM, PostgreSQL, Redis, Stripe

- Led the build of a Node.js and TypeORM API modeling institution, instructor, staff, and student domains separately.
- Built three React frontends (student, instructor, support portals) consuming the shared API.
- Implemented a Stripe-backed student wallet and payments flow, plus SCORM support for course content.
- Shipped across dev, staging, and production environments with CI/CD, automated backups, and rollback.

VerraEarth - Vetted Multi-Vendor Marketplace

Tech: Next.js, Redux Toolkit, Node.js, TypeORM, PostgreSQL, Stripe Connect

- Built customer, vendor, and admin portals against a Node.js and TypeORM API.
- Implemented Stripe Connect for split payments, with a platform fee, per-vendor transfers, and a delayed escrow-release service.
- Designed a vendor application and review flow gating storefronts behind an admin approval queue.
- Added realtime messaging, a disputes flow, and support tickets between customers and vendors.

TECHNICAL STACK

- **Frontend:** React.js, Next.js, Vue.js, Redux Toolkit
- **Backend:** Node.js, Express.js, Golang, Laravel
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Messaging:** RabbitMQ
- **Payments:** Stripe, Stripe Connect, Payoutstack
- **Infrastructure:** Docker, Docker Compose, CI/CD (GitHub Actions), Linux
- **Security:** RBAC, JWT Authentication